UNIVERSITY OF SOUTHAMPTON AICE6002W1

SEMESTER 1 EXAMINATION 2026/27

MATHEMATICS OF MACHINE LEARNING (MOCK EXAMINATION)

DURATION 120 MINS (2 HOURS)

This paper contains FOUR questions.

Mock examination. Answer ALL parts of the question in section A (40 marks) and TWO questions from section B (30 marks each)

An outline marking scheme is shown in square brackets to the right of each question.

You ***must*** write your Student ID in the box and must not write your name anywhere on the paper.

Your answers should be written in the boxes provided on the question paper. The "Additional Space" provided on the facing or adjacent page to the question may be used for corrections or extra space. Clearly indicate which question the answer corresponds to.

Scrap paper can be used - this will be discarded without being looked at.

University approved calculators MAY be used.

Student ID:

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Question	Mark	*Arithmetic checked*	*Double Marked*
A1	/40		
B2	/30		
B3	/30		
B4	/30		
Total:			

Answers

A foreign language translation dictionary (paper version) is permitted provided it contains no notes, additions or annotations.

This examination does not contain colour figures.

15 page examination paper. This paper has no handouts.

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Section A

A1

(a) Using the convention that $(\partial g/\partial \mathbf{X})_{ij} = \partial g/\partial X_{ij}$, show that for a constant matrix $\mathbf{A}$

$$\frac{\partial \operatorname{tr}(\mathbf{A}\,\mathbf{X})}{\partial \mathbf{X}} = \mathbf{A}^{\mathsf{T}} \qquad\qquad \frac{\partial \operatorname{tr}(\mathbf{X}^{\mathsf{T}}\mathbf{A}\,\mathbf{X})}{\partial \mathbf{X}} = (\mathbf{A} + \mathbf{A}^{\mathsf{T}})\,\mathbf{X}.$$

What does the second result become when $\mathbf{A}$ is symmetric?

Writing the trace in components, $\operatorname{tr}(\mathbf{AX}) = \sum_{i,j} A_{ij}X_{ji}$, so differentiating with respect to X_{lm} picks out the single term with $j = l$, $i = m$, giving A_{ml}. That is the (l, m) element of $\mathbf{A}^{\mathsf{T}}$.
For the second, $\operatorname{tr}(\mathbf{X}^{\mathsf{T}}\mathbf{AX}) = \sum_{i,j,k} X_{ji}A_{jk}X_{ki}$. The index pair (l, m) can match either factor of X:

$$\frac{\partial}{\partial X_{lm}} \sum_{i,j,k} X_{ji}A_{jk}X_{ki} = \sum_{k} A_{lk}X_{km} + \sum_{j} X_{jm}A_{jl} = (\mathbf{AX})_{lm} + (\mathbf{A}^{\mathsf{T}}\mathbf{X})_{lm}\,.$$

Hence the derivative is $(\mathbf{A} + \mathbf{A}^{\mathsf{T}})\,\mathbf{X}$, which reduces to $2\,\mathbf{A}\,\mathbf{X}$ when $\mathbf{A}$ is symmetric.

[5 marks]

(b) An inner product satisfies $\langle \boldsymbol{x}, \boldsymbol{x}\rangle \geq 0$ for all $\boldsymbol{x}$. By expanding the quadratic

$$q(t) = \langle \boldsymbol{x} + t\,\boldsymbol{y}, \boldsymbol{x} + t\,\boldsymbol{y}\rangle = A\,t^2 + 2\,B\,t + C$$

identify A, B and C, and by requiring that $q(t)$ never becomes negative deduce the Cauchy–Schwarz inequality

$$\langle \boldsymbol{x}, \boldsymbol{y}\rangle^2 \leq \langle \boldsymbol{x}, \boldsymbol{x}\rangle\,\langle \boldsymbol{y}, \boldsymbol{y}\rangle\,.$$

Expanding using bilinearity and symmetry of the inner product,

$$q(t) = \langle \boldsymbol{x}, \boldsymbol{x}\rangle + 2\,t\,\langle \boldsymbol{x}, \boldsymbol{y}\rangle + t^2\,\langle \boldsymbol{y}, \boldsymbol{y}\rangle\,,$$

so $A = \langle \boldsymbol{y}, \boldsymbol{y}\rangle$, $B = \langle \boldsymbol{x}, \boldsymbol{y}\rangle$ and $C = \langle \boldsymbol{x}, \boldsymbol{x}\rangle$.
Since $q(t) \geq 0$ for every real t, the quadratic cannot cross the axis, so it has no distinct real roots. From the standard solution $t = \left(-B \pm \sqrt{B^2 - A\,C}\right)/A$ this requires the discriminant to be non-positive,

$$B^2 - A\,C \leq 0 \qquad \Longrightarrow \qquad \langle \boldsymbol{x}, \boldsymbol{y}\rangle^2 \leq \langle \boldsymbol{x}, \boldsymbol{x}\rangle\,\langle \boldsymbol{y}, \boldsymbol{y}\rangle\,.$$

[5 marks]

Answers

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(c) Let **M** be a real symmetric matrix. Show that its eigenvalues are real, and that eigenvectors belonging to distinct eigenvalues are orthogonal.

Let $\mathbf{M}v = \lambda v$ with $v \neq 0$, allowing λ and v to be complex, and write v^* for the conjugate transpose. Then $v^*\mathbf{M}v = \lambda\, v^*v$. Taking the conjugate transpose of the whole expression and using $\mathbf{M}^* = \mathbf{M}$ (real and symmetric) gives $v^*\mathbf{M}v = \bar{\lambda}\, v^*v$. Since $v^*v = \|v\|^2 > 0$ we may divide, giving $\lambda = \bar{\lambda}$, so λ is real. For the second part let $\mathbf{M}v_1 = \lambda_1 v_1$ and $\mathbf{M}v_2 = \lambda_2 v_2$ with $\lambda_1 \neq \lambda_2$. Then

$$v_2^{\mathsf{T}}\mathbf{M}v_1 = \lambda_1\, v_2^{\mathsf{T}}v_1, \qquad v_2^{\mathsf{T}}\mathbf{M}v_1 = (\mathbf{M}v_2)^{\mathsf{T}}\, v_1 = \lambda_2\, v_2^{\mathsf{T}}v_1,$$

using symmetry in the second. Subtracting, $(\lambda_1 - \lambda_2)\; v_2^{\mathsf{T}}v_1 = 0$, and as $\lambda_1 \neq \lambda_2$ we must have $v_2^{\mathsf{T}}v_1 = 0$.

[5 marks]

(d) Using only the triangle inequality $\|\boldsymbol{x} + \boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$ and homogeneity $\|a\,\boldsymbol{x}\| = |a|\|\boldsymbol{x}\|$, show that any proper norm $\|\cdot\|$ is a convex function.

Convexity requires $\|a\boldsymbol{x} + (1-a)\boldsymbol{y}\| \leq a\|\boldsymbol{x}\| + (1-a)\|\boldsymbol{y}\|$ for all $a \in [0,1]$. The triangle inequality gives

$$\|a\,\boldsymbol{x} + (1-a)\,\boldsymbol{y}\| \leq \|a\,\boldsymbol{x}\| + \|(1-a)\,\boldsymbol{y}\|,$$

and by homogeneity, since $a \geq 0$ and $1 - a \geq 0$, this equals $a\|\boldsymbol{x}\| + (1-a)\|\boldsymbol{y}\|$, which is the required inequality.

[5 marks]

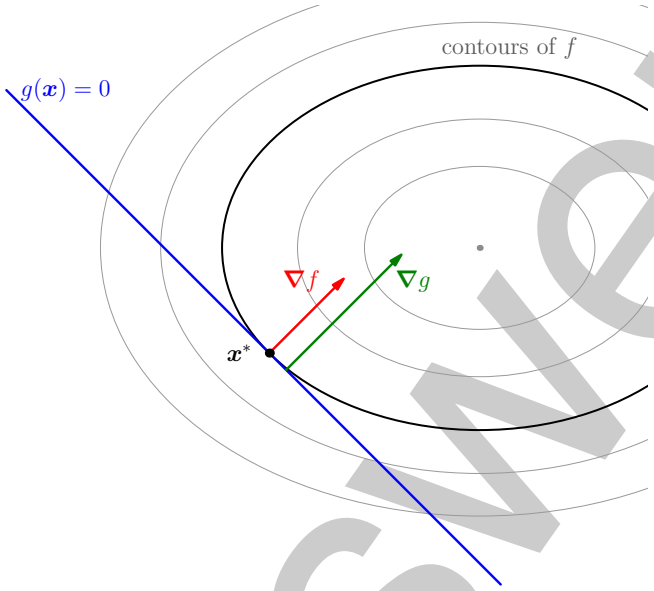
Answers

TURN OVER

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(e) We wish to maximise $f(\boldsymbol{x})$ subject to the constraint $g(\boldsymbol{x}) = 0$. Explain why at the constrained optimum the condition $\boldsymbol{\nabla} f = \lambda \boldsymbol{\nabla} g$ must hold for some scalar λ. Illustrate your answer with a sketch showing contours of f and the constraint curve.

Let δ be any small displacement that stays on the constraint surface, so that to first order $\delta^{\mathsf{T}} \nabla g = 0$; that is, the feasible displacements are exactly those orthogonal to ∇g. If x^* is a constrained maximum then f cannot increase to first order along any such direction, so $\delta^{\mathsf{T}} \nabla f = 0$ for every δ orthogonal to ∇g. A vector orthogonal to every vector orthogonal to ∇g must be parallel to ∇g, hence $\nabla f = \lambda \nabla g$. Geometrically the contour of f through x^* is tangent to the constraint curve: were they to cross, we could move along the constraint onto a higher contour.



[5 marks]

Answers

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(f) By considering the square of the integral and changing to polar coordinates, show that for $a > 0$

$$\int_{-\infty}^{\infty} \mathrm{e}^{-a\,x^2}\mathrm{d}x = \sqrt{\frac{\pi}{a}}.$$

Hence write down the normalisation constant of a normal distribution with variance σ^2.

Let I denote the integral. Writing the square as a two-dimensional integral and changing to polar coordinates, in which $x^2+y^2=r^2$ and $\mathrm{d}x\,\mathrm{d}y = r\,\mathrm{d}r\,\mathrm{d}\theta$,

$$I^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathrm{e}^{-a\left(x^2+y^2\right)}\mathrm{d}x\mathrm{d}y = \int_0^{2\pi}\int_0^{\infty} \mathrm{e}^{-ar^2}\, r\,\mathrm{d}r\,\mathrm{d}\theta.$$

The angular integral gives 2π, and substituting $u=r^2$ (so $\mathrm{d}u = 2r\,\mathrm{d}r$) the radial integral is $\int_0^{\infty}\mathrm{e}^{-au}\mathrm{d}u/2 = 1/(2a)$. Hence $I^2 = 2\pi/(2a) = \pi/a$ and $I=\sqrt{\pi/a}$. A normal distribution has $\mathrm{e}^{-x^2/(2\sigma^2)}$, i.e. $a = 1/(2\sigma^2)$, so the integral is $\sqrt{2\pi\sigma^2}$ and the normalisation constant is $1/\sqrt{2\pi\sigma^2}$.

[5 marks]

(g) We observe n successes in N independent trials, each succeeding with unknown probability p, giving the binomial likelihood $\mathbb{P}(n|p) \propto p^n\,(1-p)^{N-n}$. Show that the beta distribution

$$\mathrm{Beta}(p|a,b) = \frac{p^{a-1}\,(1-p)^{b-1}}{B(a,b)}$$

is a conjugate prior to this likelihood, and state the parameters of the posterior.

By Bayes' rule the posterior is proportional to the likelihood times the prior,

$$\mathbb{P}(p|n) \propto p^n\,(1-p)^{N-n} \times p^{a-1}\,(1-p)^{b-1} = p^{a+n-1}\,(1-p)^{b+N-n-1}\,.$$

This is again of beta form, so the posterior is $\mathrm{Beta}\,(p\,|\,a+n,\ b+N-n)$, with normalisation $1/B(a+n, b+N-n)$ since it must be a normalised density. Prior and posterior lie in the same family, which is what is meant by conjugacy: the successes are added to a and the failures to b.

[5 marks]

Answers

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(h) The Kullback–Leibler divergence between two distributions is

$$\mathrm{KL}\,(p\,\|\,q) = \sum_x p(x)\log\left(\frac{p(x)}{q(x)}\right).$$

Using Jensen's inequality, $\mathbb{E}\left[\log(Z)\right] \le \log(\mathbb{E}\left[Z\right])$, show that $\mathrm{KL}\,(p\,\|\,q) \ge 0$, and state when equality holds.

Write the divergence with the ratio inverted,

$$\mathrm{KL}\,(p\,\|\,q) = -\sum_x p(x)\log\left(\frac{q(x)}{p(x)}\right) = -\mathbb{E}\left[\log(Z)\right], \qquad Z = \frac{q(x)}{p(x)},$$

where the expectation is over x drawn from p. Jensen's inequality for the concave logarithm gives

$$\mathbb{E}\left[\log(Z)\right] \le \log\big(\mathbb{E}\left[Z\right]\big) = \log\left(\sum_x p(x)\frac{q(x)}{p(x)}\right) = \log\left(\sum_x q(x)\right) = \log(1) = 0.$$

Hence $\mathrm{KL}\,(p\,\|\,q) = -\mathbb{E}\left[\log(Z)\right] \ge 0$. Equality in Jensen's inequality for a strictly concave function requires Z to be constant, and since $\sum_x q(x) = \sum_x p(x) = 1$ that constant must be 1, so equality holds exactly when $p = q$.

[5 marks]

Answers

End of question A1

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Section B

B2 We have m data points $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_m \in \mathbb{R}^p$ with mean $\boldsymbol{\mu}$ and covariance matrix

$$\mathbf{C} = \frac{1}{m-1} \sum_{k=1}^{m} (\boldsymbol{x}_k - \boldsymbol{\mu}) (\boldsymbol{x}_k - \boldsymbol{\mu})^{\mathsf{T}}.$$

(a) The variance of the data projected onto a unit vector $\boldsymbol{v}$ is $\sigma^2(\boldsymbol{v}) = \frac{1}{m-1} \sum_{k=1}^{m} \left(\boldsymbol{v}^{\mathsf{T}} (\boldsymbol{x}_k - \boldsymbol{\mu})\right)^2$. Using a Lagrange multiplier λ for the constraint $\|\boldsymbol{v}\|^2 = 1$, show that the direction of maximum variance satisfies $\mathbf{C}\boldsymbol{v} = \lambda \boldsymbol{v}$. Show that λ is the variance in that direction, and hence say which eigenvector of $\mathbf{C}$ is the first principal component. Explain why every eigenvalue of $\mathbf{C}$ is non-negative.

Since $\left(\boldsymbol{v}^{\mathsf{T}} \boldsymbol{z}\right)^2 = \boldsymbol{v}^{\mathsf{T}} \boldsymbol{z}\, \boldsymbol{z}^{\mathsf{T}} \boldsymbol{v}$,

$$\sigma^2(\boldsymbol{v}) = \boldsymbol{v}^{\mathsf{T}} \left(\frac{1}{m-1} \sum_{k=1}^{m} (\boldsymbol{x}_k - \boldsymbol{\mu}) (\boldsymbol{x}_k - \boldsymbol{\mu})^{\mathsf{T}} \right) \boldsymbol{v} = \boldsymbol{v}^{\mathsf{T}} \mathbf{C} \boldsymbol{v}.$$

The Lagrangian is $\mathcal{L} = \boldsymbol{v}^{\mathsf{T}} \mathbf{C} \boldsymbol{v} - \lambda \left(\boldsymbol{v}^{\mathsf{T}} \boldsymbol{v} - 1\right)$, and as $\mathbf{C}$ is symmetric

$$\nabla_{\boldsymbol{v}} \mathcal{L} = 2\mathbf{C}\boldsymbol{v} - 2\lambda \boldsymbol{v} = \mathbf{0} \qquad \Longrightarrow \qquad \mathbf{C}\boldsymbol{v} = \lambda \boldsymbol{v},$$

so the stationary directions are the eigenvectors of $\mathbf{C}$. For such a direction $\sigma^2(\boldsymbol{v}) = \boldsymbol{v}^{\mathsf{T}} \mathbf{C} \boldsymbol{v} = \lambda\, \boldsymbol{v}^{\mathsf{T}} \boldsymbol{v} = \lambda$: the multiplier is the variance along $\boldsymbol{v}$. The variance is therefore largest for the eigenvector with the largest eigenvalue, which is the first principal component.
For any $\boldsymbol{v}$, $\boldsymbol{v}^{\mathsf{T}} \mathbf{C} \boldsymbol{v} = \sigma^2(\boldsymbol{v})$ is a sum of squares divided by $m-1$, so it is ≥ 0: $\mathbf{C}$ is positive semi-definite. Taking $\boldsymbol{v}$ to be a unit eigenvector gives $\lambda = \boldsymbol{v}^{\mathsf{T}} \mathbf{C} \boldsymbol{v} \geq 0$.

[10 marks]

Answers

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(b) Let $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p$ be orthonormal eigenvectors of $\mathbf{C}$ with eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$. Each data point is approximated by its projection onto the first K principal components,

$$\hat{\boldsymbol{x}}_k = \boldsymbol{\mu} + \sum_{i=1}^{K} \boldsymbol{v}_i \left(\boldsymbol{v}_i^{\mathsf{T}} \left(\boldsymbol{x}_k - \boldsymbol{\mu} \right) \right).$$

Show that the mean squared reconstruction error is

$$\frac{1}{m-1} \sum_{k=1}^{m} \| \boldsymbol{x}_k - \hat{\boldsymbol{x}}_k \|^2 = \sum_{i=K+1}^{p} \lambda_i.$$

The eigenvectors form an orthonormal basis, so $\boldsymbol{x}_k - \boldsymbol{\mu} = \sum_{i=1}^{p} \boldsymbol{v}_i \boldsymbol{v}_i^{\mathsf{T}} \left(\boldsymbol{x}_k - \boldsymbol{\mu} \right)$ and the residual is

$$\boldsymbol{x}_k - \hat{\boldsymbol{x}}_k = \sum_{i=K+1}^{p} \boldsymbol{v}_i \, a_{ki}, \qquad a_{ki} = \boldsymbol{v}_i^{\mathsf{T}} \left(\boldsymbol{x}_k - \boldsymbol{\mu} \right).$$

By orthonormality $\| \boldsymbol{x}_k - \hat{\boldsymbol{x}}_k \|^2 = \sum_{i=K+1}^{p} a_{ki}^2$. Averaging over the data,

$$\frac{1}{m-1} \sum_{k=1}^{m} \| \boldsymbol{x}_k - \hat{\boldsymbol{x}}_k \|^2 = \sum_{i=K+1}^{p} \frac{1}{m-1} \sum_{k=1}^{m} a_{ki}^2 = \sum_{i=K+1}^{p} \boldsymbol{v}_i^{\mathsf{T}} \mathbf{C} \boldsymbol{v}_i = \sum_{i=K+1}^{p} \lambda_i,$$

using part (a) for the variance along each $\boldsymbol{v}_i$. Keeping the directions of largest variance therefore also minimises the reconstruction error.

[5 marks]

Answers

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(c) In LoRA (low-rank adaptation), a pretrained $m \times n$ weight matrix $\mathbf{W}_0$ is frozen and a network is fine-tuned by learning $\mathbf{W} = \mathbf{W}_0 + \mathbf{AB}$, where $\mathbf{A}$ is $m \times K$, $\mathbf{B}$ is $K \times n$ and $K \ll m, n$.

(i) Show that $\mathbf{AB}$ has rank at most K, and compare the number of trainable parameters with full fine-tuning when $m = n = 4096$ and $K = 8$.

(ii) Let $\mathbf{G} = \partial L/\partial \mathbf{W}$ be the gradient of the loss, $G_{ij} = \partial L/\partial W_{ij}$. Show that $\partial L/\partial \mathbf{A} = \mathbf{G}\mathbf{B}^{\mathsf{T}}$ and $\partial L/\partial \mathbf{B} = \mathbf{A}^{\mathsf{T}}\mathbf{G}$.

(iii) LoRA initialises $\mathbf{B} = \mathbf{0}$ and $\mathbf{A}$ at random. Using (ii), explain why training still makes progress, and why initialising both factors at zero would fail.

(i) Each column of AB is A times a column of B, so lies in the span of the K columns of A: the rank is at most K. LoRA trains $(m+n)K = 8192 \times 8 = 65\,536$ parameters against $mn = 16\,777\,216$, about 0.4%.
(ii) $W_{ij} = (W_0)_{ij} + \sum_k A_{ik}B_{kj}$, so by the chain rule

$$\frac{\partial L}{\partial A_{ik}} = \sum_j G_{ij}B_{kj} = \left(\mathbf{G}\mathbf{B}^{\mathsf{T}}\right)_{ik} \qquad \frac{\partial L}{\partial B_{kj}} = \sum_i G_{ij}A_{ik} = \left(\mathbf{A}^{\mathsf{T}}\mathbf{G}\right)_{kj}.$$

(iii) With $\mathbf{B} = \mathbf{0}$ the network starts exactly at the pretrained $\mathbf{W}_0$. The gradient for A is zero, but the gradient for B is $\mathbf{A}^{\mathsf{T}}\mathbf{G}$, which is non-zero for random A, so B moves on the first step and A starts moving after it. If both were zero both gradients would vanish: $\mathbf{A} = \mathbf{B} = \mathbf{0}$ is a stationary point (a saddle) and gradient descent would never leave it.

[8 marks]

(d) LoRA rests on the hypothesis that the change a new task needs has low "intrinsic rank". In an experiment, all weights of a 1024×1024 matrix are fine-tuned and the resulting update $\Delta\mathbf{W}$ has singular values $6, 3, 2, 1$, with the remaining 1020 singular values all equal to 0.1. Using the Eckart–Young theorem, which you may quote, find the squared Frobenius error of the best rank-K approximation for $K = 1, 2, 3, 4$, the fraction of $\|\Delta\mathbf{W}\|_F^2$ captured with $K = 4$, and the smallest K that captures 90%. Comment on what this says about the hypothesis.

Eckart–Young: the best rank-K approximation is the truncated SVD, with $\|\Delta\mathbf{W} - \Delta\mathbf{W}_K\|_F^2 = \sum_{i>K} s_i^2$. Here $\|\Delta\mathbf{W}\|_F^2 = 36 + 9 + 4 + 1 + 1020 \times 0.01 = 60.2$, so the squared errors are

$$K = 1:\ 24.2 \qquad K = 2:\ 15.2 \qquad K = 3:\ 11.2 \qquad K = 4:\ 10.2.$$

With $K = 4$ the fraction captured is $50/60.2 \approx 83\%$. For 90% we need $0.9 \times 60.2 = 54.18$, i.e. 4.18 more from components worth 0.01 each: 418 more, so $K = 422$.
Four directions carry most of the structure, but the remaining 17% is spread flat over a thousand directions, and capturing it costs a hundred times more rank. A flat tail looks like noise (for instance from stochastic gradients); if it is, a small K discards it and may even regularise the fit, supporting the hypothesis. If the tail matters for the task, small K loses it. The spectrum alone cannot decide which: one would compare validation performance against K.

[7 marks]

End of question B2

TURN OVER

Answers

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B3 A learning machine is trained on a data set $\mathcal{D}$ of m examples drawn independently from a distribution $\mu(\boldsymbol{x}, y)$, giving a predictor $\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_\mathcal{D})$ with generalisation loss

$$L_G(\boldsymbol{\theta}_\mathcal{D}) = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x}, y) \left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_\mathcal{D}) - y\right)^2 .$$

The *mean machine* is $\hat{f}_m(\boldsymbol{x}) = \mathbb{E}_\mathcal{D}\left[\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_\mathcal{D})\right]$.

(a) Show that the expected generalisation loss $\bar{L}_G = \mathbb{E}_\mathcal{D}[L_G(\boldsymbol{\theta}_\mathcal{D})]$ can be written as $\bar{L}_G = B + V$, where

$$B = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x}, y) \left(\hat{f}_m(\boldsymbol{x}) - y\right)^2 \qquad V = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x}, y)\, \mathbb{E}_\mathcal{D}\left[\left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_\mathcal{D}) - \hat{f}_m(\boldsymbol{x})\right)^2\right].$$

Explain in words what each term measures, and how each typically changes as the model is made more complex.

The sum over (x, y) does not depend on $\mathcal{D}$, so the expectation can be taken inside. Adding and subtracting the mean machine,

$$\mathbb{E}_\mathcal{D}\left[\left(\hat{f} - y\right)^2\right] = \mathbb{E}_\mathcal{D}\left[\left(\hat{f} - \hat{f}_m\right)^2\right] + \left(\hat{f}_m - y\right)^2 + 2\,\mathbb{E}_\mathcal{D}\left[\hat{f} - \hat{f}_m\right]\left(\hat{f}_m - y\right),$$

where $\hat{f} = \hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_\mathcal{D})$ and we used that $\hat{f}_m(\boldsymbol{x}) - y$ does not depend on $\mathcal{D}$. The cross term vanishes because $\mathbb{E}_\mathcal{D}\left[\hat{f} - \hat{f}_m\right] = \hat{f}_m - \hat{f}_m = 0$. Weighting by $\mu(\boldsymbol{x}, y)$ and summing gives $\bar{L}_G = B + V$.
B, the bias, is the loss of the mean machine: how well the model class can fit even when the luck of the particular data set is averaged away (it also includes the irreducible noise in y). V, the variance, measures how much the learned predictor changes from one data set to another. A more complex model usually has smaller bias, since it can represent the target better, but larger variance, since it can also fit the noise of each data set: the bias–variance dilemma. More data reduces V.

[10 marks]

Answers

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(b) Consider the simplest learning problem: estimating the mean θ of a distribution with variance σ^2 from m independent samples with sample mean $\bar{x}$, using the *shrinkage* estimator $\hat{\theta} = c\,\bar{x}$ with $0 \le c \le 1$. (Shrinkage is used in statistics to reduce the sample estimate error in systems with high variance. It deliberately underestimates the mean.) Find $\mathbb{E}\left[\hat{\theta}\right]$ and $\mathrm{Var}(\hat{\theta})$, and hence show that

$$\mathbb{E}\left[\left(\hat{\theta}-\theta\right)^2\right] = (1-c)^2\,\theta^2 + c^2\,\frac{\sigma^2}{m}.$$

$\mathbb{E}\left[\bar{x}\right] = \theta$ and, the samples being independent, $\mathrm{Var}(\bar{x}) = \sigma^2/m$. Hence $\mathbb{E}\left[\hat{\theta}\right] = c\,\theta$ and $\mathrm{Var}(\hat{\theta}) = c^2\sigma^2/m$. As in part (a), adding and subtracting $\mathbb{E}\left[\hat{\theta}\right]$ and noting that the cross term has zero mean,

$$\mathbb{E}\left[\left(\hat{\theta}-\theta\right)^2\right] = \left(\mathbb{E}\left[\hat{\theta}\right]-\theta\right)^2 + \mathrm{Var}(\hat{\theta}) = (1-c)^2\theta^2 + c^2\frac{\sigma^2}{m}:$$

a squared bias plus a variance.

[5 marks]

(c) An ensemble predicts with the average $\bar{f}_M(\boldsymbol{x}) = \frac{1}{M}\sum_{i=1}^{M} f_i(\boldsymbol{x})$ of M predictors trained with different randomness (resampled data, random initialisation). At a fixed $\boldsymbol{x}$, suppose that each $f_i(\boldsymbol{x})$ has mean $\hat{f}_m(\boldsymbol{x})$ and variance s^2 over this randomness, and that every pair has correlation ρ. Show that

$$\mathrm{Var}\left(\bar{f}_M(\boldsymbol{x})\right) = \rho\,s^2 + \frac{1-\rho}{M}\,s^2.$$

Using part (a), say which term of the expected loss an ensemble can reduce and what limits the reduction.

$\mathrm{Cov}\left(f_i, f_j\right) = s^2$ **if** $i = j$ **and** ρs^2 **otherwise, so**

$$\mathrm{Var}\left(\bar{f}_M\right) = \frac{1}{M^2}\sum_{i,j}\mathrm{Cov}\left(f_i, f_j\right) = \frac{1}{M^2}\left(Ms^2 + M(M-1)\rho s^2\right) = \rho s^2 + \frac{1-\rho}{M}s^2.$$

The mean is unchanged, $\mathbb{E}\left[\bar{f}_M\right] = \hat{f}_m$, so the ensemble has the same mean machine and the same bias B. Only the variance term V is reduced, and however large M is it cannot fall below ρs^2: the correlation between members sets the floor.

[8 marks]

Answers

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(d) Random forests choose each split of each tree from a random subset of the features, which makes the individual trees worse but less correlated. Suppose bagged trees have $s^2 = 1$ and $\rho = 0.6$, while random-forest trees have $s^2 = 1.5$ and $\rho = 0.2$. Compare the variance terms for $M = 1$ and $M = 100$, and find the ensemble size above which the random forest wins. Deep ensembles train several neural networks on the *same* data from different random initialisations. Use the formula of part (c) to explain why this helps, and say how the spread of the members' predictions can be used.

For $M = 1$ the variances are 1 (bagging) and 1.5 (forest): a single forest tree is worse. For $M = 100$: bagging $0.6 + 0.4/100 = 0.604$, forest $0.3 + 1.2/100 = 0.312$. They are equal when $0.6 + 0.4/M = 0.3 + 1.2/M$, i.e. $M = 8/3$, so the forest wins for $M \geq 3$. For large ensembles the floor ρs^2 dominates, so lowering the correlation matters more than the quality of each member.
A network's loss is non-convex, so different initialisations end in different minima, giving members with low ρ even on the same data, and the ensemble removes much of the variance. The spread of the members' predictions at x estimates this variance term, so it can be reported as an uncertainty: it is typically larger away from the training data, flagging predictions that should not be trusted.

[7 marks]

Answers

End of question B3

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B4 In Bayesian inference we want expectations under the posterior $\pi(\boldsymbol{\theta}) = \mathbb{P}(\mathcal{D}|\boldsymbol{\theta})\,\mathbb{P}(\boldsymbol{\theta})\,/\mathbb{P}(\mathcal{D})$ of a parameter vector $\boldsymbol{\theta} \in \mathbb{R}^d$.

(a) Explain why computing a posterior expectation by numerical integration on a grid becomes infeasible as d grows. Show that the Monte Carlo estimate $\hat{g} = \frac{1}{n}\sum_{i=1}^{n} g(\boldsymbol{\theta}_i)$, from n independent samples drawn from π, is unbiased and has a standard error proportional to $1/\sqrt{n}$ whatever the value of d.

A grid with k points along each axis has k^d points, which grows exponentially with d; even the evidence $\mathbb{P}(\mathcal{D})$ is such an integral. For the Monte Carlo estimate, $\mathbb{E}\left[\hat{g}\right] = \frac{1}{n}\sum_i \mathbb{E}\left[g(\boldsymbol{\theta}_i)\right] = \mathbb{E}\left[g\right]$, so it is unbiased, and by independence

$$\mathrm{Var}(\hat{g}) = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}\left(g(\boldsymbol{\theta}_i)\right) = \frac{\mathrm{Var}(g)}{n},$$

a standard error $\sqrt{\mathrm{Var}(g)}/\sqrt{n}$. The dimension enters only through $\mathrm{Var}(g)$, not through the rate.

[5 marks]

(b) A Markov chain with transition density $W(\boldsymbol{\theta} \to \boldsymbol{\theta}')$ satisfies *detailed balance* with respect to π if $W(\boldsymbol{\theta} \to \boldsymbol{\theta}')\,\pi(\boldsymbol{\theta}) = W(\boldsymbol{\theta}' \to \boldsymbol{\theta})\,\pi(\boldsymbol{\theta}')$.

(i) Show that π is then a stationary distribution of the chain.

(ii) The Metropolis–Hastings algorithm proposes $\boldsymbol{\theta}'$ from a proposal density $q(\boldsymbol{\theta}'|\boldsymbol{\theta})$ and accepts it with probability

$$A(\boldsymbol{\theta} \to \boldsymbol{\theta}') = \min\left(1, \frac{\pi(\boldsymbol{\theta}')\,q(\boldsymbol{\theta}|\boldsymbol{\theta}')}{\pi(\boldsymbol{\theta})\,q(\boldsymbol{\theta}'|\boldsymbol{\theta})}\right).$$

Show that the resulting chain satisfies detailed balance, and explain why the evidence $\mathbb{P}(\mathcal{D})$ is not needed.

(i) Integrating detailed balance over $\boldsymbol{\theta}$, and using that the chain must go somewhere, $\int W(\boldsymbol{\theta}' \to \boldsymbol{\theta})\,\mathrm{d}\boldsymbol{\theta} = 1$,

$$\int W(\boldsymbol{\theta} \to \boldsymbol{\theta}')\,\pi(\boldsymbol{\theta})\,\mathrm{d}\boldsymbol{\theta} = \pi(\boldsymbol{\theta}') \int W(\boldsymbol{\theta}' \to \boldsymbol{\theta})\,\mathrm{d}\boldsymbol{\theta} = \pi(\boldsymbol{\theta}').$$

The left side is the distribution after one step starting from π, so π is unchanged by the chain.
(ii) For $\boldsymbol{\theta}' \neq \boldsymbol{\theta}$, $W(\boldsymbol{\theta} \to \boldsymbol{\theta}') = q(\boldsymbol{\theta}'|\boldsymbol{\theta})\,A(\boldsymbol{\theta} \to \boldsymbol{\theta}')$. Let r be the ratio inside the minimum and suppose $r \leq 1$; then $A(\boldsymbol{\theta} \to \boldsymbol{\theta}') = r$ while the reverse ratio is $1/r \geq 1$, so $A(\boldsymbol{\theta}' \to \boldsymbol{\theta}) = 1$. Hence

$$W(\boldsymbol{\theta} \to \boldsymbol{\theta}')\,\pi(\boldsymbol{\theta}) = q(\boldsymbol{\theta}'|\boldsymbol{\theta})\,\frac{\pi(\boldsymbol{\theta}')\,q(\boldsymbol{\theta}|\boldsymbol{\theta}')}{\pi(\boldsymbol{\theta})\,q(\boldsymbol{\theta}'|\boldsymbol{\theta})}\,\pi(\boldsymbol{\theta}) = q(\boldsymbol{\theta}|\boldsymbol{\theta}')\,\pi(\boldsymbol{\theta}') = W(\boldsymbol{\theta}' \to \boldsymbol{\theta})\,\pi(\boldsymbol{\theta}').$$

The case $r > 1$ is the same with the roles exchanged, and a rejected move ($\boldsymbol{\theta}' = \boldsymbol{\theta}$) satisfies detailed balance trivially. The acceptance depends on π only through the ratio $\pi(\boldsymbol{\theta}')/\pi(\boldsymbol{\theta})$, in which $\mathbb{P}(\mathcal{D})$ cancels: we need only likelihood times prior.

[10 marks]

TURN OVER

Answers

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(c) Let (x, y) be jointly Gaussian with zero means, unit variances and correlation r, so that $\pi(x, y) \propto \exp\left(-\left(x^2 - 2rxy + y^2\right) / \left(2(1 - r^2)\right)\right)$. By completing the square, show that $x \,|\, y \sim \mathcal{N}\left(r\,y,\ 1 - r^2\right)$. A Gibbs sampler alternately draws x from $p(x|y)$ and y from $p(y|x)$. Show that a Gibbs update of x is a Metropolis–Hastings step, with proposal $q(x'|x, y) = p(x'|y)$, whose acceptance probability is always 1.

As a function of x at fixed y, $x^2 - 2rxy = (x - ry)^2 - r^2y^2$, so

$$\pi(x, y) \propto \exp\left(-\frac{(x - ry)^2}{2(1 - r^2)}\right) \times (\textbf{terms in } y \textbf{ only}),$$

which is a normal density in x with mean ry and variance $1 - r^2$.
Write $\pi(x, y) = p(x|y)\,p(y)$. The Metropolis–Hastings ratio for moving $x \to x'$ at fixed y is

$$\frac{\pi(x', y)\,q(x|x', y)}{\pi(x, y)\,q(x'|x, y)} = \frac{p(x'|y)\,p(y)\;p(x|y)}{p(x|y)\,p(y)\;p(x'|y)} = 1,$$

so every proposal is accepted: Gibbs sampling is Metropolis–Hastings with a proposal that matches the target exactly in the coordinate being updated.

[7 marks]

(d) For the Gibbs sampler of part (c), show that the values of x after successive sweeps obey $x_{t+1} = r^2 x_t + \xi_t$, where ξ_t is independent noise of variance $1 - r^4$. You may assume that at stationarity $\mathrm{Cov}\,(x_t, x_{t+k}) = \rho^k$ with $\rho = r^2$. Show that for large n the mean $\bar{x}$ of n successive values has variance approximately $\frac{1}{n}\frac{1+\rho}{1-\rho}$, and hence give the effective sample size. Evaluate it for $r = 0.95$, and comment on what this means for MCMC on strongly correlated posteriors.

One sweep draws $y_{t+1} = r x_t + \sqrt{1 - r^2}\,\zeta$ and then $x_{t+1} = r y_{t+1} + \sqrt{1 - r^2}\,\zeta'$, with ζ, ζ' independent standard normals. Substituting, $x_{t+1} = r^2 x_t + \xi_t$ with $\xi_t = r\sqrt{1 - r^2}\,\zeta + \sqrt{1 - r^2}\,\zeta'$, whose variance is $r^2(1 - r^2) + (1 - r^2) = 1 - r^4$.
With unit variance at stationarity,

$$\mathrm{Var}(\bar{x}) = \frac{1}{n^2}\sum_{s,t=1}^{n} \rho^{|s-t|} \approx \frac{1}{n}\left(1 + 2\sum_{k=1}^{\infty}\rho^k\right) = \frac{1}{n}\left(1 + \frac{2\rho}{1 - \rho}\right) = \frac{1}{n}\,\frac{1 + \rho}{1 - \rho},$$

neglecting end effects for large n. Independent samples would give $1/n$, so the effective sample size is $n_{\text{eff}} = n\,(1 - \rho)/(1 + \rho)$. For $r = 0.95$, $\rho = 0.9025$ and $(1 + \rho)/(1 - \rho) = 1.9025/0.0975 \approx 19.5$: $10\,000$ sweeps are worth only about 510 independent samples.
Strong posterior correlations, common when parameters are poorly identified, make coordinate-wise and random-walk samplers mix slowly. This is why MCMC results should report the effective sample size rather than the chain length, and why one reparametrises to remove correlations, updates correlated parameters together, or uses gradient-based samplers.

[8 marks]

End of question B4

Answers

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END OF PAPER

This page contains no examination material

Answers